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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 09/28/2004 |
| **PI Name:** | Wang, Hai Ph.D. | | |
| **Project Title:** | Soot Formation in Purely-Curved Premixed Flames and Laminar Flame Speeds of Soot-Forming Flames | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | BIOTECHNOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Bio-nanotechnology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 19716-3140 | **Congressional District:** | 1 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 99-HEDS-04 |
| **Start Date:** | 12/01/2000 | **End Date:** | 11/30/2004 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 3 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Sung, Chih-Jen ( Case Western Reserve University ) | | |
| **Grant/Contract No.:** | NCC3-843 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | This joint program aims to gain fundamental understanding of the structure and dynamics of sooting premixed flames in simple, well-defined, one-dimensional, burner-supported, cylindrical and spherical flow fields. The proposed program consists of the following four major components: (1) experimental determination of the fundamental sooting limits in the absence of conductive heat loss to the burner; (2) experimental determination of the fundamental laminar flame speed of soot-forming flames, (3) experimental mapping of sooting structure, and (4) detailed modeling of sooting flame propagation and soot formation. The work in components 1 and 2 will demonstrate a novel experimental approach under microgravity and generate fundamental flame and soot data that are suitable for rigorous comparison with detailed numerical simulation. This experimental approach eliminates the ambiguities associated with the boundary conditions in previous soot laminar premixed flame experiments and simulation. Combined with components 3 and 4 of the proposed work, we shall develop a soot research platform which is unique and complementary to the existing soot research tools and facilities. | | |

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| | To study rich, freely-propagating, sooting premix flames by microgravity experiments and numerical simulation.<br><br>This experiment is the first time that sooting flames are studied without the added complications of flame stretch or heat loss to the burner. The results of the study will yield fundamental insight into minimizing particulate pollutant emission from combustion. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Flame soot is a carcinogen and also contributes to global warming. Current work aims to further our understanding of sooting premixed flames, and improve current sooting models. This will allow better predictions of soot formation in practical combustion systems. |
| **Task Progress:** | - Compact Rainbow Schlieren Deflectometry (RSD) optical system was built and tested under 1-g conditions. Optical system was shown to accurately determine the location of the premixed-flame surface even under heavily sooting conditions, where direct imaging is not possible. RSD also provided the flame thickness and 2D temperature distribution as a function of time. System was later integrated into the microgravity rig.<br>- Experimental microgravity rig was built and tested. Different burner configurations (spherical/cylindrical), sizes and porosities were tested under microgravity conditions for optimal results. The final 3/8" diam spherical burner was determined to provide the most symmetric and repeatable flames.<br><br>- A total of 140 microgravity drops have been performed to date.<br><br>- Premixed burning velocities of ethylene/air flames were studied as a function of burner flowrate and equivalence ratio. The spherical flame configuration was proven as an accurate method for determining the burning velocities of very rich sooting flames. The fundamental unstretched burning velocities were obtained for the previously unmeasured equivalence ratios of 3.0, 3.5, 4.0, and 4.5. Burning velocity values as low as 2.0 cm/s were measured using this method.<br><br>- Experimental measurements were compared with a complex gas-phase chemistry model. Measured velocity values for all equivalence ratios studied were significantly lower.<br><br>- Experiment is ongoing. Current focus is on the effect of nitrogen dilution on the sooting characteristics and burning velocities of rich ethylene/air flames.<br><br>- Experimental measurements were made for the evolution of particle size distribution in ethylene flames. The measurements emphasize the role of flame temperature on the characteristics of the particle size distribution. This information is critical to the development of a predictive model of soot formation. |
| **Bibliography Type:** | Description: (Last Updated: 01/11/2021) |
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| **Articles in Peer-reviewed Journals** | Ibarreta AF, Sung CJ. "Flame temperature and location measurements of sooting premixed Bunsen flames by rainbow schlieren deflectometry." Appl Opt. 2005 Jun 10;44(17):3565-75. https://doi.org/10.1364/AO.44.003565 ; PMID: 16007857 , Jun-2005 |
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| **Papers from Meeting Proceedings** | Ibarreta AF, Sung CJ. "Temperature Measurements in Sooting Premixed Flames Using Rainbow Schlieren Deflectometry." 42nd AIAA Aerospace Sciences. Meeting and Exhibit. Reno, Nevada, 5-8 January, 2004.<br>42nd AIAA Aerospace Sciences. Meeting and Exhibit. Reno, Nevada, 5-8 January, 2004. , Jan-2004 |
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